University of Exeter Medical School

Assignment Brief - Coursework 2

Module Title:	Making a Difference with Health Data	**Module Convenor:**	Thomas Monks
Module Code:	HPDM097	**Summative/Formative**	Summative
Module Credits:	30	**Assignment Type:**	Group
% of Module Mark:	50%	**Date Set:**	13/01/2026

Date Due:	**27/03/2026**		

1 Required Task

The vast majority of Discrete-Event Simulation (DES) models used to analyse health problems are published in academic journal articles as text descriptions only. That is, the computer model - the code - is not available to anyone but the original study authors; however, a description of the model logic and data are provided in writing. The recreation of computer models from these natural language descriptions is non-trivial and requires a specialist with DES training.

- In this assignment, you will work in a group (size 2 or 3) to select an academic journal article reporting a DES model applied to a health problem. You will then conduct a research study to test if it is possible to recreate the simulation model (or a simplified version of it) in python and 'simpy'.
- You are permitted to manually write the simulation code yourselves in python, and/or conduct experiments with Large Language Models (LLM; e.g. ChatGPT, Claude, or Gemini) to investigate their ability to recreate models in python.
- You are required to produce a report documenting your research study, results, and conclusions. You must also provide a technical appendix (one or more Jupyter notebooks) containing the Python code and all prompts given to a LLM.

2 Assignment components

Please submit a **single** .zip file to ELE containing the following:

1. A **report** of your research study to recreate a DES model (see Section 4.1 for requirements)
2. A **technical appendix**: a well formatted and annotated Jupyter notebook(s) of the full simulation model and any prompts to an AI tool. (see Section 4.2 for requirements).
3. **Evidence of group working**:
   - As this is a group assignment, we want to see evidence that the group operated in a manner similar to what would be expected in a professional environment. Therefore, we are requiring that you submit a portfolio of documents to showcase this.
     - Meeting agendas (x3)
     - Meeting minutes; including attendees and actions assigned to individuals (x3)
     - Action plan with milestones

3 Advice on the recreation of DES models

Figure 1 illustrates a general recommended approach for how a model can be recreated manually or by using an LLM. In both cases, it is recommended to take an iterative approach to model development. In each iteration, additional detail is added to the simulation model. This is followed by a series of tests that are conducted to check if the new model logic is implemented as expected and also to test that the new logic does not break anything already implemented. Document and record everything you do with care.

3.1 General advice on getting started

It is recommended that groups follow this process to get started:

1. Carefully read a paper and produce a list or diagram of all of the activities (e.g., treatment), resources, and routing that patients follow in the model.
2. List what data are reported in the paper (such as parameters for statistical distributions) and if any assumptions or simplifications are needed.
3. Before writing or generating any code, discuss and plan simplifications in the model logic, and a potential design for the model code in 'simpy' (e.g. classes, functions, Experiment classes, how to run multiple replications)

3.2 How to add detail to the model

Building the model in one go will be difficult. It is recommended that the groups add detail step by step. For example:

- Step 1: Write code for generating one or more types of patient arrivals to the health care system.
- Step 2: Test the arrival processes. If all tests pass, proceed to step 3.

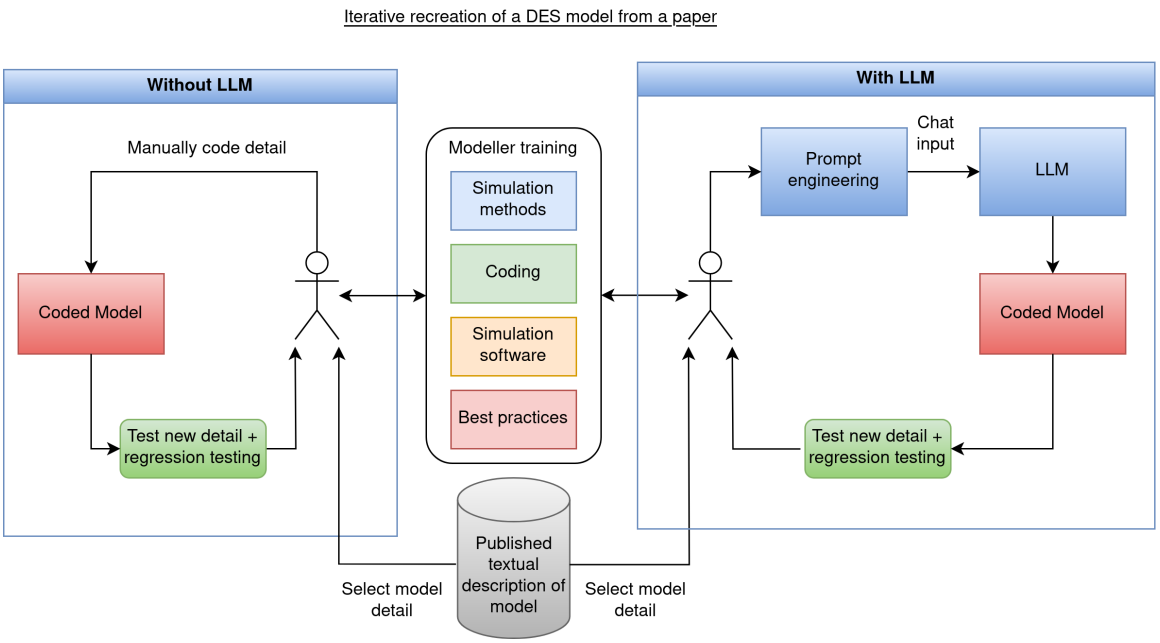

Figure 1: A recommended iterative approach to recreating models manually or with the aid of an LLM.

- Step 3: Write code where patients undergo treatment or queue and take resources.
- Step 4: Test that the treatment process works as expected. Retest the arrival process.

3.3 How to use a Large Language Model in your research

LLMs are hard to use for specialist data science applications. If you choose to use one then your research should focus on **'prompt engineering'**. A *prompt* is the query string that you provide to the chat interface of the LLM. Current LLM's cannot cope with all of the text from an academic paper and the natural language in the paper may be too ambiguous for an LLM to generate code that you might expect. Groups are advised to do the following:

- Limit the size of *prompts*, by designing ones that generate and modify code incrementally. For example, if your model has multiple activities, add these in one at a time using multiple prompt and testing cycles.
- Be specific about python libraries that you wish to use.
- Be consistent with terminology used within a single prompt or in subsequent prompts. For example, if you have initially used the term 'treatment time' to define a patient activity in a ward then you should always refer to 'treatment time' and avoid alternative terms such as 'length of stay'.
- Take extra care to provide prompts that define logic clearly. For example define steps taken in a function in a numbered list or use clear terminology for if/else logic.

Groups are reminded that LLM's can make mistakes and may be trained on code that may contain obvious or subtle bugs. Test your code incrementally. If you find bugs you could attempt to identify the problem in your original prompt and re-engineer it; prompt the LLM to fix the bug; or fix the bug manually. **This is a research project so keep a detailed log of what you have done and think critically about what types of prompts work and what types don't work for recreating DES models.**

See sections 4.1 and 4.2 for advice on what to document and how to report it.

3.4 Where to find academic journal articles

Groups are free to choose their own article. To help, several example journal articles are provided and groups are allowed to use these for their project.

- Penn, M. L., Monks, T., Kazmierska, A. A., & Alkoheji, M. R. A. R. (2019). Towards generic modelling of hospital wards: Reuse and redevelopment of simple models. Journal of Simulation, 14(2), 107–118. `https://doi.org/10.1080/17477778.2019.1664264`
- Lahr, Maarten M.H. et al. (2013) A Simulation-based Approach for Improving Utilization of Thrombolysis in Acute Brain Infarction. Medical Care 51(12):p 1101-1105, DOI: 10.1097/MLR.0b013e3182a3e505 `https://doi.org/10.1097/MLR.0b013e3182a3e505`
- J D Griffiths, M Jones, M S Read & J E Williams (2010) A simulation model of bed-occupancy in a critical care unit, Journal of Simulation, 4:1, 52-59, DOI: `https://doi.org/10.1057/jos.2009.22`
- Monks, T. et al. , "A modelling tool for capacity planning in acute and community stroke services," BMC Health Serv Res 16, 530 (2016). `https://doi.org/10.1186/s12913-016-1789-4`

Penn et al. includes two DES models of hosptial wards (one general one Intensive Care). The models were created in a DES package called Simul8. The Simul8 model for the general ward and example parameters are included in an online appendix (in an Excel file). The queuing in the general ward model has to deal with the added complexity that NHS wards separate male and female patients.

Lahr et al. used a commercial DES package called PlantSim to investigate acute stroke provision in the Netherlands. It has a non-typical approach to simulation where a fixed number of patient journeys are simulated as opposed to using multiple replications. No queuing is included. You can find detailed data in the online appendix. No computer model is available to the public.

Griffiths et al. is a relatively simple and classical model of a Critical Care Unit in the UK. It was coded in VBA and has never been made available. It has a detailed description of the model and most of its parameters; although some assumptions will have to be made about distributions. For example, where empirical data was used instead of a statistical distribution.

Monks et al, used a commercial DES package called Simul8 to build a simple generic model of acute stroke and rehabilitation capacity. It has a clear logic diagram and good documentation of parameters. An option for simplification is to focus only on the acute stroke service. The Simul8 computer model is not available to the public.

Alternatively you may wish to search the literature for a suitable **healthcare** DES model. It is recommended that you pick a model with low to medium levels of complexity. For example, models that have relatively simple queuing processes, resource usage, and are limited in scope. To help your search you may wish to consult one of the following sources:

- The **Healthcare Applications** track in Winter Simulation Conference archives. `https://informs-sim.org/`
- The **Journal of Simulation** archives `https://www.tandfonline.com/journals/tjsm20`

For any model selected, groups are also permitted to simplify the model reported in the academic paper. For example,

- Remove complex booking/scheduling rules in favour of simpler queuing,
- Reduce the number of activities included in a model,
- Model a single type of patient instead of multiple types of patients (e.g. a general stroke patient versus patients with severe, moderate and mild stroke),
- Choose to use a simple statistical distribution (e.g. Log Normal) instead of a complex phase type distribution.

All simplifications must be documented and justified in the report. **At a minimum your model design should either have one type of patient and multiple activities; or multiple types of patient and one activity.**

4 The report and technical appendix

4.1 The report

The report is a write up of your research study attempting to recreate a DES model. It should include an abstract (or summary), introduction, methods, results, discussion, conclusions and **references**. The report should not exceed 2000 words including the abstract. Please ensure that it is written in good quality English and is free from spelling mistakes.

4.1.1 Abstract

An abstract should be no more than 200 words. It should summarise objectives, methods, results and conclusions.

4.1.2 Introduction

The introduction should list your research questions, cite the journal article, **briefly** introduce the design of the DES model that you are recreating (for example, a description and diagrams of model logic), and describe any assumptions, or simplifications that you are required to make due to inadequate reporting in the academic article.

4.1.3 Methods

Methods should describe the process you used for recreating the model (e.g what LLM(s) were used, the process for prompt engineering, types of tests used), what data you recorded/captured at each iteration of the model (e.g. number and functions of tests), how you will measure success/failure (e.g. what tests should the final model pass to be considered a recreation of your model design?), and how you handled any unexpected problems you encountered. Cite any journal articles, websites, and sources that you used to design your study and methods.

4.1.4 Results

Results should be brief and report the data and success/failure criteria described in your methods. Text, tables and charts may be used. Results may include, but are not limited to:

- a description of the final model, how to use it, example output from the model, and if you were successful or not in your aims to recreate the design in the introduction.
- report how many iterations were needed to recreate the model.
- number of tests failed at each iteration and rework

Results tables will not count towards your word count.

4.1.5 Discussion

The discussion provides an opportunity for groups to reflect on the project. Groups may consider:

- what were the problems in reporting in the paper that prevented, limited or aided their recreation of the model? What were the positive aspects of the paper that aided recreation?
- what could the authors of the article have done to improve their model documentation to aid recreation?
- giving more context to their prompt engineering and provide examples of prompts that worked and prompts that did not work to generate acceptable code. (Note: example prompts will not count towards total word count).
- what were the challenges with using an LLM and how were these overcome?
- what are the limitations of their results and method?
- what would they do or record differently if they could re-design their methods and start again?
- if there is any recent literature in recreating models (of any type) that has similar or different findings.

4.1.6 Conclusion

A brief paragraph with concluding remarks to the study.

4.1.7 References

References should use the Vancouver (numbered) reference style. References do not count towards the 2000 word limit.

4.2 Technical Appendix

The appendix should include a well formatted and annotated Jupyter notebook of the full simulation model. The notebook should demonstrate either how the model is used, or its current deficiencies if it is not possible to fully recreate the model.

If an LLM has been used in an iterative manner to support recreation of the model then each iteration should have its own notebook. These notebooks should contain the aim of the prompt, the full prompt given to the LLM, the code (and any text) generated by the LLM, testing of the code, and tester notes. Please name the notebooks so that the order of prompts is obvious (e.g. `01_iter.ipynb`, `02_iter.ipynb` ...).

It must be possible to run your Jupyter notebooks. Please error check them and provide details of a virtual environment to use.

4.3 Marking criteria

Marking is split into two components

4.3.1 Evidence of group working (10%)

Please note that the evidence of group working will not be marked for content but compliance only. Therefore, if you submit all the documents and they are fit for purpose you will receive the full 10% for this component.

4.3.2 Simulation research study (90%)

A detailed breakdown of these criteria is available at the end of the document.

1. The design and execution of the research study.
2. The quality of the report.
3. The quality of the technical appendix.

5 Marking Scheme

The design and execution of the research study

- **0 - 49**: Absent or limited evidence of appropriate design and an unclear method for conducting the research. The simulation model design does not meet the minimum complexity specification.
- **50 - 59**: Limited but mostly appropriate study design and method for conducting the research.
- **60 - 69**: Generally appropriate study design and method for recreating an appropriately complex simulation model. A rationale for the study methods is presented, but has some areas of ambiguity or easily overcome limitations.
- **70 - 85**: An appropriate and study design and clear method for recreating an appropriately complex simulation model. A strong rationale for the study methods is presented, all but the most challenging aspects of appropriate research methodology are included.

- **86 - 100**: A research standard study design that is informed by the literature and attempts to manage potential method limitations seen elsewhere.

The quality of the report

- **0 - 49**: Absent or poor quality report that writes up the study, containing a large number of grammatical and typographical errors. The report exceeds the word limit by more than 10%.
- **50 - 59**: Borderline passable report that writes up the study. The report contains multiple grammatical and typographical errors.
- **60 - 69**: Generally good report that attempts to take account of all requirements specified. The report may contain grammatical and typographical errors.
- **70 - 85**: An excellent report that takes account of all requirements specified. The report has minimal grammatical and typographical errors.
- **86 - 100**: Publication quality write-up of the research that accounts for all aspects of the specification.

The quality of the technical appendix

- **0 - 49**: Absent or unintuitive technical appendices. The code in the appendix cannot be run by a marker.
- **50 - 59**: Borderline passable technical appendices. Demonstration of limited understanding of how to create Jupyter notebooks, organise code, and follow best practice. The code in the appendix can be run by a marker with some manual intervention.
- **60 - 69**: Generally good technical appendices. Demonstration of understanding of how to create Jupyter notebooks, organise code, and follow most aspects of coding best practice. The code in the appendix can be run by a marker without the need for manual intervention, but may not follow best practices.
- **70 - 85**: Excellent technical appendices that demonstrate a solid understanding of how to create Jupyter notebooks, organise code, and follow best practice. The code in the appendix is easy to run and follows best practice.
- **86 - 100**: Publication quality Jupyter notebooks that are easily understood, and navigated by someone who did not create them. All code dependencies are simple to manage and the notebooks can be reused to recreate the results of the study.

6 Submission

Your assignment must be submitted to ELE by the stated deadline. For instructions on how to do this, please see the Online Submission Student Handbook (`https://www.exeter.ac.uk/students/infopoints/yourinfopointservices/assessments/`).

If you submit your work late, but are within one hour of the deadline, a penalty of 5% will be deducted from your mark. For any work submitted more than an hour late your mark will be capped at 50% (up to 24hrs after the original deadline; after which a score of zero will be awarded). This is unless you have valid mitigation (please visit the Mitigation SharePoint page for more information.)

Please ensure that your name DOES NOT appear anywhere in the document, but that your personal ID number is included. You should submit your work as a **.zip** file. The naming of your file must follow the format below:

- `ID_number_ModuleCode_assessment_name_AC_year`
- E.g. `91000000_CSC3019_written_review_2025-26`

Please note that submitting an incorrect file, or submitting the correct file to the wrong location, will not be considered as grounds for mitigation. Late penalties will be applied if you submit the correct file to the right location after the submission deadline.

In submitting this you are declaring the work is your own independent piece of work, not produced in collusion with a fellow student or plagiarized from a fellow student, or a website, or a textbook, or any other information source. You must demonstrate good referencing practice and ensure you have sufficiently paraphrased all sources of information. The direct copying of AI-generated content is also included under plagiarism, misrepresentation and contract cheating under definitions and offences in TQA Manual Chapter 12.3. During the submission process you will be asked to check a box, that confirms you have adhered to the university's academic conduct policy and the expectations on use of Generative AI (GenAI) in your assessment brief. Failure to do so may result in being referred to an academic misconduct panel which has the power to grant penalties based on specific criteria. For more details, please revisit the Academic Honesty and Plagiarism information on ELE.

7 Use of AI in Assessment

This assessment falls under the category of AI-Assisted in the University's Guidance on use of GenAI in Assessment. This is because using AI tools in specific ways can assist with your learning and will not inhibit fair assessment of your achievement of the module's intended learning outcomes. This means: You may use GenAI tools ethically and responsibly to assist in the development of an assessment in accordance with the boxes checked below.

- ☑ To develop ideas
- ☑ To assist with research or information gathering
- ☑ To help you understand key concepts and theories
- ☑ To identify trends and themes as part of data analysis
- ☑ To provide feedback on a draft
- ☑ To improve the plan or structure of my assessment
- ☑ To generate images, figures or diagrams
- ☑ To proofread and correct spelling or grammar errors
- ☑ To format citations or references
- ☐ To translate material into English in line with guidance from your module lead. Unless this box is checked, you must never use AI translate more than a word or short phrase into English.

- ☑ OTHER: To support debugging of Python code you have written.
- ☑ OTHER: To improve the quality of Python docstrings
- ☑ OTHER: To improve code quality
- ☑ OTHER [To generate the code for the simulation model as part of the research]

When writing your assessment, you must never use AI tools:

1. For uses other than those represented by checked boxes in the list above.
2. To translate more than a word or short phrase into English unless agreed above.
3. To upload sensitive or identifying material to an AI tool.
4. To present material that has been generated by AI as your own work or the work of someone else.

When submitting your assessment, you must:

1. Check the box during the submission process, that confirms you have adhered to the university's academic conduct policy and the expectations on use of GenAI in your assessment brief.
2. Treat the AI tool like a citation from any other source.
3. Include a list of all AI prompts and hyperlinks to their output with your references, at the end of your work. You do not need to include the outputs themselves, just the links.
4. Retain the full outputs generated by the prompts you have used during your assignment. These outputs should be accessible at the hyperlinks which you have submitted with your assignment. You may be asked to produce this material in the event of an academic conduct inquiry.

8 Support for Students with English as a second language

International students and those who have English as a second language can make use of the English Language Skills Development programme. This programme provides face-to-face workshops and courses as well as one-to-one assignment writing tutorials and an extensive range of online resources via their Guided Independent Learning site. Find out more by browsing the timetables: `https://www.exeter.ac.uk/into/englishlanguage/howtoparticipate/timetablessupport` or emailing `insessional@exeter.ac.uk`